Summary of the Benders Decomposition

Large Scale Optimization

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1. Examples of MILP

A very simple example

FIXED CHARGE PROBLEMS

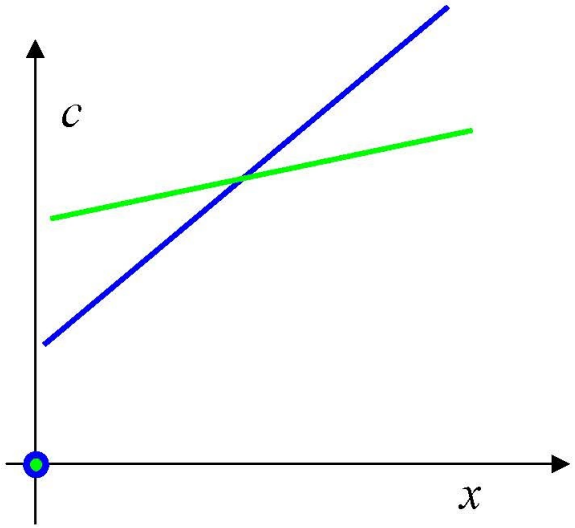
A factory plans to produce two types of products A and B. The total production is guaranteed to be sold. A resource is necessary (120 units)

	A	B
Benefit per unit	200	500
# resources per unit	3	6
Subsidy	2000	1000

$$3x_A + 6x_B \leq 120, \quad x_A \geq 0, \; x_B \geq 0.$$

$$c_A(x_A) = \begin{cases} 200x_A + 2000 & \text{si } x_A > 0 \\ 0 & \text{si } x_A = 0 \end{cases}$$

$$c_B(x_B) = \begin{cases} 500x_B + 1000 & \text{si } x_B > 0 \\ 0 & \text{si } x_B = 0 \end{cases}$$



A very simple example

PROBLEM'S REFORMULATION

$$
\begin{aligned}
Max_{\ x} \quad & c_A(x_A) + c_B(x_B) \\
& 3x_A + 6x_B \leq 120 \\
& x_A \geq 0,\ x_B \geq 0, x_A,\ x_B \in Z
\end{aligned}
$$

If $d_A, d_B \in \{0, 1\}$:

$$
\boxed{
\begin{array}{llll}
x_A > 0 & \Rightarrow d_A = 1 & x_B > 0 & \Rightarrow d_B = 1 \\
x_A = 0 & \Rightarrow d_A = 0 & x_B = 0 & \Rightarrow d_B = 0
\end{array}
}
$$

$$c_A(x_A) = 200x_A + 2000d_A$$

$$c_B(x_B) = 500x_B + 1000d_B$$

A very simple example

$$C = \{ x_A, x_B \in Z \mid 3x_A + 6x_B \leq 120,\ x_A \geq 0,\ x_B \geq 0 \}$$

An upper bound for x_A in C is $\hat{x}_A = 40$

A positive lower bound for x_A in C is $\underline{x}_A = 1$

d_A binaria

$$d_A \leq x_A \leq 40 d_A. \quad \rightarrow \quad \begin{array}{l} x_A > 0 \ \Rightarrow\ d_A = 1 \\ x_A = 0 \ \Rightarrow\ d_A = 0 \end{array}$$

$$\boxed{\begin{array}{ll} Max_{\ x,\ d} & 200x_A + 2000d_A + 500x_B + 1000d_B \\ & 3x_A + 6x_B \leq 120 \\ & (x_A \geq 0,\ x_B \geq 0) \\ & d_A \leq x_A \leq 40d_A, \\ & d_B \leq x_B \leq 20d_B \\ & 0 \leq d_A,\ d_B \leq 1 \\ & x_A,\ x_B, d_A, d_B \in Z \end{array}}$$

The multicommodity network design problem

The problem of network design can be formulated :

$$
\begin{array}{ll}
Min_{\,x,y} & \sum_{\ell \in K} c^{\ell\top} x^{\ell} + f^{\top} y \\
(A) & B x^{\ell} = t^{\ell}, \;\; \ell \in K \\
(B) & x_a^{\ell} \leq \rho\, y_a, \;\; \ell \in K, \; a \in \hat{A} \\
(C) & x^{\ell} \geq 0 \\
 & y \in \{0,1\}^{|\hat{A}|}
\end{array}
$$

and in scalar notation:

$$
\begin{array}{lll}
Min_{\,x,y} & \sum_{\ell \in K} \sum_{a \in A} c_a^{\ell} x_a^{\ell} + \sum_{a \in \hat{A}} f_a y_a & \\
 & \sum_{r \in I(i)} x_{r,i}^{\ell} \; - \; \sum_{s \in E(i)} x_{i,s}^{\ell} \; = \; t_i^{\ell}, & i \in N, \; \ell \in K \\
 & x_a^{\ell} \leq \rho\, y_a, & a \in \hat{A}, \; \ell \in K \\
 & x_a^{\ell} \geq 0, & a \in A, \; \ell \in K \\
 & y_a \in \{0,1\}, & a \in \hat{A}
\end{array}
$$

2. Formulation of the BD method

MILP Formulation

Reference MILP formulation

$$\begin{array}{lll} \phi^* = \text{Min}_{x,y} & f^\top y + c^\top x & \\ \text{s.t.} & Dx = d - Fy & \mid \; u \\ \text{(MILP)} & x \geq 0, y \in Y & \end{array}$$

Linear problems for each $y \in Y$

$$\begin{array}{lll} v(y) = \text{Min}_x & c^\top x & \\ \text{s.t.} & Dx = d - Fy & \\ \text{(P)} & x \geq 0 & \end{array}$$

$$\begin{array}{lll} v(y) = \text{Max}_u & (d - Fy)^\top u & \\ \text{(D)} \quad \text{s.t.} & D^\top u \leq c & \end{array}$$

Alternative formulation

$$\phi^* = \text{Min}_{y \in Y_0} \; \phi(y)$$

$$\phi(y) \triangleq f^\top y + v(y)$$

$$Y_0 \triangleq \{y \in Y \mid v(y) < \infty\}$$

Dual Polyhedron

$$D^\top u \leq c \;\rightarrow \begin{cases} \hat{u}^1, ..., \hat{u}^p \text{ vertices} \\ \vec{w}^1, ..., \vec{w}^q \text{ extreme directions} \end{cases}$$

Master Problem

$$\text{(MP)} \qquad \begin{aligned} \varphi(I,J) = \text{Min}_{z,y} \quad & z \\ & z \geq f^\top y + (d - Fy)^\top \hat{u}_i, && i \in I \\ & 0 \geq (d - Fy)^\top \vec{w}_j, && j \in J \\ & I \subseteq \tilde{I} = \{1, 2, ..., p\},\; J \subseteq \tilde{J} = \{1, 2, ..., q\} \end{aligned}$$

Subproblem

Evaluate $\phi(y) \triangleq f^\top y + v(y)$ by solving either (P) or (D)

Essential properties

1. $\phi^* = \varphi(\tilde{I}, \tilde{J})$
2. If $\left\{ \begin{array}{c} \bar{y} \notin Y^* \\ I \subset \tilde{I} \\ J \subset \tilde{J} \end{array} \right\} \Rightarrow \phi(\bar{y}) > \phi^* = \mathsf{Min}_{y \in Y_0} \phi(y) \geq \varphi(I, J)$
3. $\mathsf{relgap} = \dfrac{\phi(y) - \varphi(I, J)}{|\varphi(I, J)| + \epsilon}$

Algorithm outline

Find $y^0 \in Y$, set $\varphi^0 = -\infty$, $\ell = 0$, $I^0 = J^0 = \emptyset$

<u>At ℓ-th iteration:</u>

1. Evaluate the Subproblem: $\phi(y^\ell)$
2. Stopping criteria: If relgap $= \dfrac{\phi(y^\ell) - \varphi^\ell}{|\varphi^\ell| + \epsilon} \leq \varepsilon \rightarrow$ STOP
3.
   - Finite Dual: $v(y^\ell) < +\infty$; new vertex $\hat{u}_\ell$; $I^{\ell+1} = I^\ell \cup \{\ell\}$.
   - Unbounded Dual: $v(y^\ell) = +\infty$; new extr.direction $\vec{w}_j$; $J^{\ell+1} = J^\ell \cup \{j\}$
4. Solve MP: $\varphi^{\ell+1} = \varphi(I^{\ell+1}, J^{\ell+1})$
5. $\ell \longleftarrow \ell + 1$